BE CAPSTONE PROJECT · 2026–27

# Battery State Estimator

*An Echo State Network alternative to the Extended Kalman Filter and Coulomb Counting*

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01 — PROBLEM & OBJECTIVES

# Charge is easy to measure badly, hard to estimate well.

Accurate SOC and SOH estimation keeps Battery Management Systems safe and efficient in EVs and storage systems. The classical methods each fail in a different way.

## Coulomb Counting

Open-loop current integration. Sensor offsets accumulate, so the estimate drifts over time.

## Extended Kalman Filter

Needs parameter identification and recalibration as cells age. Matrix maths is costly on small MCUs, and Q and R need manual tuning.

### OBJECTIVES

- Design an ESN (reservoir computing) estimator for SOC and SOH
- Build a 2-RC electro-thermal battery simulator
- Benchmark against EKF, Coulomb Counting and VFF-RLS
- Optimise inference with sparse maths and fixed-point arithmetic
- Validate in C99 on STM32 and Verilog RTL on an FPGA
- Build a live web dashboard and publish the research

*Goal: a lightweight, data-driven estimator that reads only voltage, current and temperature.*

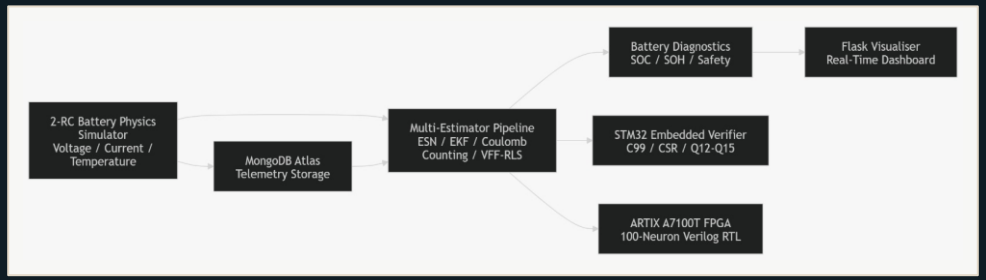
02 — SYSTEM ARCHITECTURE

# One pipeline, from simulated cell to silicon.

## EXECUTION FLOW




## ARCHITECTURE




- Simulator: 2-RC cell model with DST, US06 and FUDS drive cycles
- Telemetry: V, I and T stored in MongoDB Atlas and served by Flask
- Estimators: ESN runs alongside EKF, Coulomb Counting and VFF-RLS
- Diagnostics: thermal runaway, sensor dropout and micro-short alerts
- Verifiers: C99 on STM32 and 100-neuron Verilog on an FPGA

# A physics ground truth, and a reservoir that learns it.

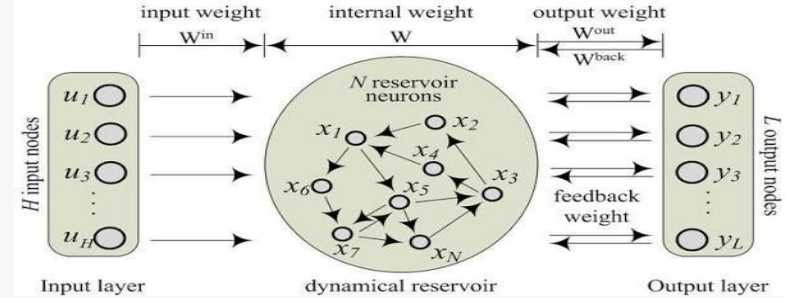
## Battery physics engine

- 2-RC equivalent circuit: OCV(SOC), ohmic R0 and two polarisation branches (R1C1, R2C2)
- Thermal balance with Arrhenius temperature scaling
- Ground-truth SOC and SOH plus noisy V, I and T sensor readings
- Fault injector for thermal runaway, dropouts and micro-shorts

*No parameter identification and no integration drift: the model learns the cell from data.*

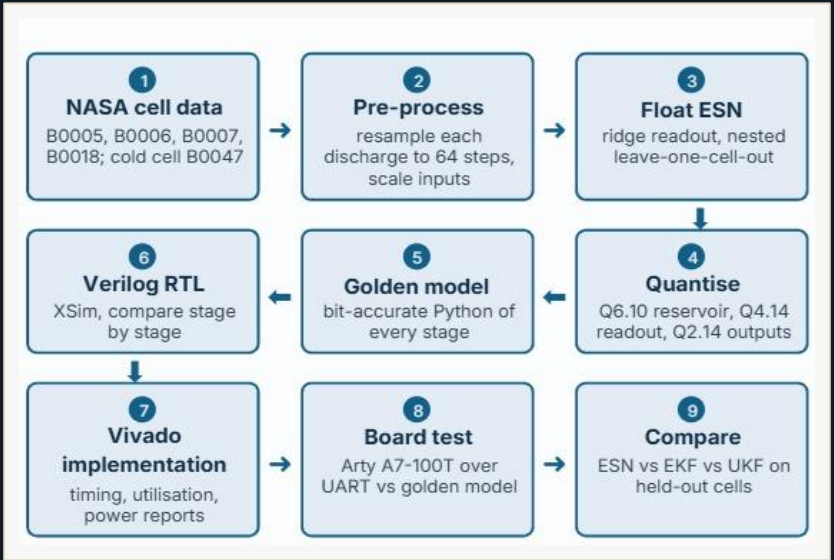
## Echo State Network estimator

- Fixed sparse reservoir (leaky integrator, spectral radius 0.95); only the readout is trained, by ridge regression
- Online VFF-RLS adaptation tracks ageing; UKF is a benchmark



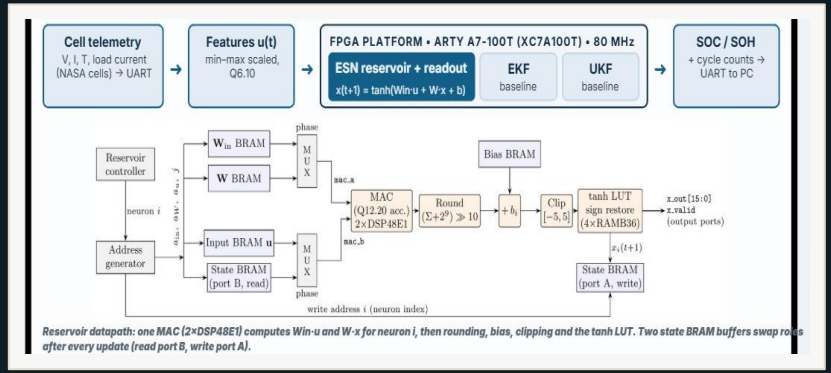
# From NASA cell data to a running board.

## VERIFICATION WORKFLOW



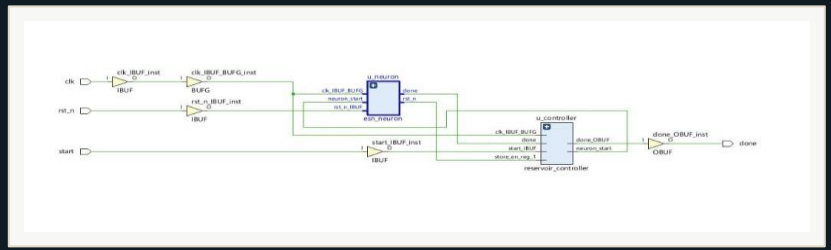

STM32 C99: sparse CSR maths, Q12/Q15 fixed point, tanh LUT. FPGA: 100 neurons, BRAM state, checked against a golden model.

## PLATFORM AND RESERVOIR DATAPATH




Reservoir datapath: one MAC (2×DSP48E1) computes Win·u and W·x for neuron i, then rounding, bias, clipping and the tanh LUT. Two state BRAM buffers swap roles after every update (read port B, write port A).

## SYNTHESISED RTL TOP LEVEL



05 — RESULTS

# Verified in software and in hardware.

| 6.7× | 200/200 | <1.5% | 85% |
|---|---|---|---|
| faster inference with CSR | bit-exact RTL matches | SOC RMSE | memory saved |

## WHAT WAS TESTED

- Physics telemetry under dynamic drive cycles
- ESN vs EKF vs Coulomb Counting, SOC RMSE under 1.5%
- Live fault injection triggers safety diagnostics
- CSR sparse inference: 6.7× speedup on C99
- 100-neuron RTL: 200/200 bit-exact matches
- 46 automated tests, 100% pass, CI on GitHub Actions

# Small enough for the edge, honest about what is next.

## Applications

- EV battery management systems
- Grid and renewable energy storage
- Low-power microcontroller diagnostics
- Hardware-in-the-loop research

## Current limits

- Telemetry is simulated; no physical cells yet
- Accuracy depends on diverse training drive cycles
- Single-cell scope only

## Future scope

- Benchtop tests with real Li-ion cells
- Flash STM32 Nucleo with CAN bus
- Multi-cell packs with active balancing
- FPGA bitstream and power profiling

**Publications:** a review paper on classical observers and reservoir computing, and an experimental paper on edge-based sparse reservoir computing. Both drafted, targeting IEEE / Q3 Scopus journals.